

Q-1- what are the different protocols you observe at the following layers of the protocol stack?

Ans→	Application Layer	Transport layer	Network layer
	MDNS	UDP	ARP
	HTTP	TCP	ICMPV6
	SSDP		IGMPV3
	DHCP		
	DNS		

Q-2- what is the total amount of data being received for the following two cases?

A) when you access :- <http://neverssl.com>

B) when you access :- <https://www.cornell.edu>

ip.addr==10.85.17.19						
No.	Time	Source	Destination	Protocol	Length	Info
1312	15.465927095	10.85.17.19	255.255.255.255	UDP	867	47209 → 29810 Len=825
3739	45.569314312	10.85.17.19	255.255.255.255	UDP	867	47209 → 29810 Len=825
413	5.429738891	10.85.17.19	255.255.255.255	UDP	868	47209 → 29810 Len=826
2153	25.500036425	10.85.17.19	255.255.255.255	UDP	868	47209 → 29810 Len=826
2912	35.534306722	10.85.17.19	255.255.255.255	UDP	868	47209 → 29810 Len=826
5496	65.639501682	10.85.17.19	255.255.255.255	UDP	868	47209 → 29810 Len=826
6326	75.674323473	10.85.17.19	255.255.255.255	UDP	868	47209 → 29810 Len=826
4680	55.604466191	10.85.17.19	255.255.255.255	UDP	869	47209 → 29810 Len=827

Q-3 Answer the following when you access the site <http://neverssl.com>

A) How many HTTP GET requests do you observe?
List down the GET requests.

Ans → Two HTTP GET requests observed.

GET requests :-

Time	Source	Destination	Protocol	Length	Info
566967.897392209	10.85.16.193	34.223.124.45	HTTP	142	GET/HTTP/1.1
598571.276595089	10.85.16.193	91.189.91.96	HTTP	134	GET/HTTP/1.1

B) What version of HTTP the server is running?

GET/HTTP/1.1

*wlo1

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http.request.method == "GET"						
No.	Time	Source	Destination	Protocol	Length	Info
5669	67.897392289	10.85.16.193	34.223.124.45	HTTP	142	GET / HTTP/1.1
5985	71.276595089	10.85.16.193	91.189.91.96	HTTP	154	GET / HTTP/1.1

c) For each of the HTTP Get request find out:
i) The total number of TCP segments being received

Ans → one TCP segment received

ii) The total amount of data being received in the corresponding HTTP response message.

Ans → 4261 bytes.

Wireshark network traffic analysis interface showing a packet capture on interface wlo1. The main display area shows a list of captured packets with columns for No., Time, Source, Destination, Protocol, and Length. The selected packet (No. 5783) is a TCP segment from 10.05.16.193 to 34.223.124.45, Seq=1, Ack=1, Win=64256, Len=0, TSval=3729545268, TSecr=3532113175. The packet details pane shows the structure of the TCP segment, including the header and options (PSH, ACK). The packet bytes pane shows the raw data of the segment.

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*wlo1

tcp

No.	Time	Source	Destination	Protocol	Length	Info
5668	67.89712984	10.05.16.193	34.223.124.45	TCP	60	51654 → 80 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=3729545268 TSecr=3532113175
5784	69.228257300	10.05.16.193	34.223.124.45	TCP	60	51654 → 80 [ACK] Seq=77 Ack=4262 Win=60032 Len=0 TSval=3729546591 TSecr=3532113175
5845	69.597352631	10.05.16.193	34.223.124.45	TCP	60	51654 → 80 [ACK] Seq=78 Ack=4263 Win=62464 Len=0 TSval=3729566926 TSecr=3532113472
5785	69.229816258	10.05.16.193	34.223.124.45	TCP	60	51654 → 80 [FIN, ACK] Seq=77 Ack=4262 Win=62464 Len=0 TSval=3729546591 TSecr=3532113175
5416	64.517987849	10.05.16.193	34.223.124.45	TCP	74	51654 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=3729541880 TSecr=0 WS=128
5804	71.276368855	10.05.16.193	91.189.91.96	TCP	60	52064 → 80 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=3413952797 TSecr=1828042714
6029	71.585158394	10.05.16.193	91.189.91.96	TCP	60	52064 → 80 [ACK] Seq=80 Ack=80 Win=64256 Len=0 TSval=3413953106 TSecr=1928043024
5952	70.966865879	10.05.16.193	91.189.91.96	TCP	74	52064 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=3413952797 TSecr=0 WS=128
5772	69.121848278	34.223.124.45	10.05.16.193	TCP	60	80 → 51654 [ACK] Seq=1 Ack=77 Win=29312 Len=0 TSval=3532113049 TSecr=3729546147
5844	69.597296516	34.223.124.45	10.05.16.193	TCP	60	80 → 51654 [FIN, ACK] Seq=4262 Ack=78 Win=29312 Len=0 TSval=3532113472 TSecr=3729546591
5667	67.897052833	34.223.124.45	10.05.16.193	TCP	74	80 → 51654 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 SACK_PERM TSval=3532111799 TSecr=3729544930 WS=128
6027	71.585098256	91.189.91.96	10.05.16.193	TCP	60	80 → 52064 [FIN, ACK] Seq=80 Ack=89 Win=65536 Len=0 TSval=1928043824 TSecr=3413952797
6078	71.585098348	91.189.91.96	10.05.16.193	TCP	60	80 → 52064 [RST, ACK] Seq=81 Ack=89 Win=65536 Len=0 TSval=1928043824 TSecr=3413952797
6061	71.888624871	91.189.91.96	10.05.16.193	TCP	60	80 → 52064 [RST] Seq=80 Win=0 Len=0
5983	71.276251881	91.189.91.96	10.05.16.193	TCP	74	80 → 52064 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 SACK_PERM TSval=1928042714 TSecr=3413952797 WS=16384
5669	67.897392209	10.05.16.193	34.223.124.45	HTTP	142	GET / HTTP/1.1
5985	71.276305989	10.05.16.193	91.189.91.96	HTTP	134	GET / HTTP/1.1
5783	69.228259144	34.223.124.45	10.05.16.193	HTTP	4327	HTTP/1.1 200 OK (text/html)
6026	71.585098159	91.189.91.96	10.05.16.193	HTTP	145	HTTP/1.1 204 No Content
5732	68.704270868	10.05.16.193	34.223.124.45	TCP	142	[TCP Retransmission] 51654 → 80 [PSH, ACK] Seq=1 Ack=1 Win=64256 Len=76 TSval=3729546147 TSecr=3532111799
5483	65.519348880	10.05.16.193	34.223.124.45	TCP	74	[TCP Retransmission] 51654 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=3729542882 TSecr=0 WS=128
5553	66.543331884	10.05.16.193	34.223.124.45	TCP	74	[TCP Retransmission] 51654 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=3729543906 TSecr=0 WS=128
5636	67.567226454	10.05.16.193	34.223.124.45	TCP	74	[TCP Retransmission] 51654 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=3729544930 TSecr=0 WS=128

Frame 5783: 4327 bytes on wire (34616 bits), 4327 bytes captured (34616 bits) on interface wlo1, id 0

Ethernet II, Src: ASIXElectron_35:2d:a8 (f8:e4:3b:35:2d:a8), Dst: 58:82:b5:d9:eb:cc (58:82:b5:d9:eb:cc)

Internet Protocol Version 4, Src: 34.223.124.45, Dst: 10.05.16.193

Transmission Control Protocol, Src Port: 80, Dst Port: 51654, Seq: 1, Ack: 77, Len: 4261

Source Port: 80

Destination Port: 51654

[Stream index: 6]

Conversation completeness: Complete, WITH_DATA (31)

[TCP Segment Len: 4261]

Sequence Number: 1 (relative sequence number)

Sequence Number (raw): 1797796701

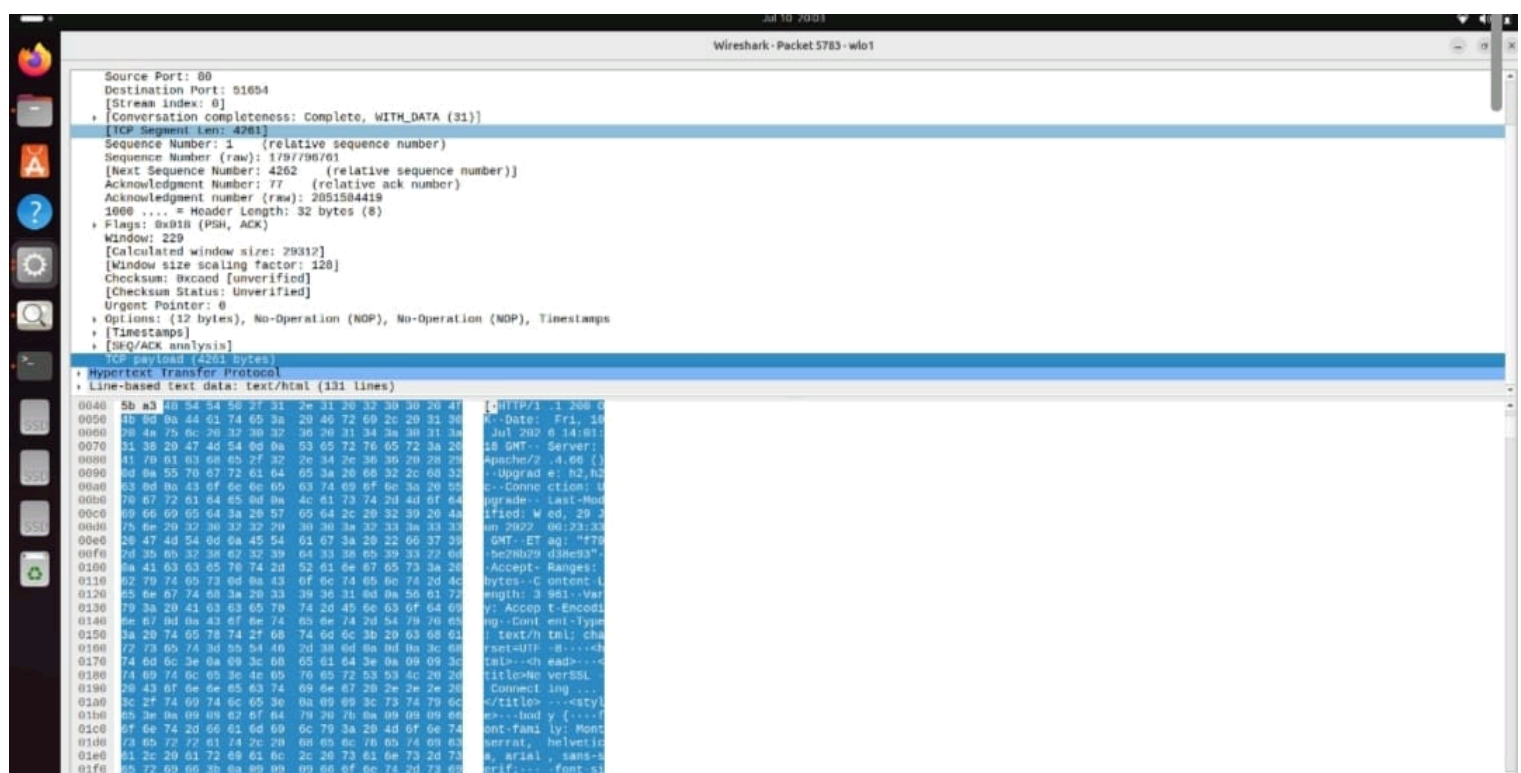
[Next Sequence Number: 4262 (relative sequence number)]

Acknowledgment Number: 77 (relative ack number)

Acknowledgment number (raw): 2851584419

1000 = Header Length: 32 bytes (8)

Flags: 0x0118 (PSH, ACK)



1281 bytes.

D) Identify TCP three way handshake used to establish the connection with the <http://neverssl.com>. Specify SYN, SYN-ACK, ACK packets and write the sequence number for each of the three accesses.

Ans → • SYN packet:-

Source IP:- 10.85.17.18 (Client)

Destination IP:- 34.223.124.45 (Server)

Protocol info:- 57022 → 80 [SYN]

• SYN-ACK packet:-

Source IP:- 34.223.124.45 (Server)

Destination IP:- 10.85.17.18 (Client)

Protocol info:- 80 → 57022 [SYN, ACK]

Sequence Number:- 0

(ACK=1)

• ACK packet:-

Source IP:- 10.85.17.18 (Client)

Destination IP:- 34.223.124.45 (Server)

Protocol Info 57022 → 80 [ACK]

Sequence Number:- 1

(ACK=1)

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No.	Time	Source	Destination	Protocol	Length	Info
5668	67.897129284	10.85.16.193	34.223.124.45	TCP	66	51654 → 80 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=3729545268 TSecr=3532111799
5784	69.228257300	10.85.16.193	34.223.124.45	TCP	66	51654 → 80 [ACK] Seq=77 Ack=4262 Win=60032 Len=0 TSval=3729546591 TSecr=3532113175
5845	69.557352631	10.85.16.193	34.223.124.45	TCP	66	51654 → 80 [ACK] Seq=78 Ack=4263 Win=62464 Len=0 TSval=3729546926 TSecr=3532113472
5785	69.229616258	10.85.16.193	34.223.124.45	TCP	66	51654 → 80 [FIN, ACK] Seq=77 Ack=4262 Win=62464 Len=0 TSval=3729546591 TSecr=3532113175
5416	64.517967849	10.85.16.193	34.223.124.45	TCP	74	51654 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=3729541880 TSecr=0 WS=128
5884	71.276300055	10.85.16.193	91.189.91.96	TCP	66	52064 → 80 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=3413952797 TSecr=1928042714
6029	71.585158394	10.85.16.193	91.189.91.96	TCP	66	52064 → 80 [ACK] Seq=80 Ack=80 Win=64256 Len=0 TSval=3413953106 TSecr=1928043024
5952	70.966865879	10.85.16.193	91.189.91.96	TCP	74	52064 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=3413952487 TSecr=0 WS=128
5772	69.121840270	34.223.124.45	10.85.16.193	TCP	66	80 → 51654 [ACK] Seq=1 Ack=77 Win=29312 Len=0 TSval=3532113049 TSecr=3729546147
5844	69.557296516	34.223.124.45	10.85.16.193	TCP	66	80 → 51654 [FIN, ACK] Seq=4262 Ack=78 Win=29312 Len=0 TSval=3532113472 TSecr=3729546591
5667	67.897052833	34.223.124.45	10.85.16.193	TCP	74	80 → 51654 [SYN, ACK] Seq=0 Ack=1 Win=26847 Len=0 MSS=1460 SACK_PERM TSval=3532111799 TSecr=3729544930 WS=128
6027	71.585098256	91.189.91.96	10.85.16.193	TCP	66	80 → 52064 [FIN, ACK] Seq=80 Ack=89 Win=65536 Len=0 TSval=1928043024 TSecr=3413952797
6028	71.585098348	91.189.91.96	10.85.16.193	TCP	66	80 → 52064 [RST, ACK] Seq=81 Ack=89 Win=65536 Len=0 TSval=1928043024 TSecr=3413952797
6041	71.608624871	91.189.91.96	10.85.16.193	TCP	66	80 → 52064 [RST] Seq=80 Win=0 Len=0
5983	71.276751881	91.189.91.96	10.85.16.193	TCP	74	80 → 52064 [SYN, ACK] Seq=0 Ack=1 Win=65536 Len=0 MSS=1460 SACK_PERM TSval=1928042714 TSecr=3413952487 WS=16384
5669	67.897392209	10.85.16.193	34.223.124.45	HTTP	142	GET / HTTP/1.1
5986	71.276595689	10.85.16.193	91.189.91.96	HTTP	154	GET / HTTP/1.1
5737	67.275232774	10.85.16.193	34.223.124.45	HTTP	4327	HTTP/1.1 200 OK (text/html)
6020	71.585098150	91.189.91.96	10.85.16.193	HTTP	145	HTTP/1.1 204 No Content
5732	66.704270808	10.85.16.193	34.223.124.45	TCP	142	[TCP Retransmission] 51654 → 80 [PSH, ACK] Seq=1 Ack=1 Win=64256 Len=78 TSval=3729546147 TSecr=3532111799
5483	65.519348889	10.85.16.193	34.223.124.45	TCP	74	[TCP Retransmission] 51654 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=3729542882 TSecr=0 WS=128
5553	65.54331884	10.85.16.193	34.223.124.45	TCP	74	[TCP Retransmission] 51654 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=3729543986 TSecr=0 WS=128
5636	67.567226454	10.85.16.193	34.223.124.45	TCP	74	[TCP Retransmission] 51654 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=3729544930 TSecr=0 WS=128

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Ethernet II, Src: ASIXElectron_35:2d:a8 (f8:e4:3b:35:2d:a8), Dst: 98:82:b5:d9:eb:cc (98:82:b5:d9:eb:cc)
Internet Protocol Version 4, Src: 34.223.124.45, Dst: 10.85.16.193
Transmission Control Protocol, Src Port: 80, Dst Port: 51654, Seq: 1, Ack: 77, Len: 4261
Source Port: 80
Destination Port: 51654
[Stream index: 0]
[Conversation completeness: Complete, WITH_DATA (31)]
[TCP Segment Len: 4261]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 1797796761
[Next Sequence Number: 4262 (relative sequence number)]
Acknowledgment Number: 77 (relative ack number)
Acknowledgment Number (raw): 2851584419
1000 = Header Length: 32 bytes (8)
Flags: 0x010 (PSH, ACK)